WASHINGTON UNIVERSITY IN ST. LOUIS

OLIN BUSINESS SCHOOL

DAT 5564: Database Design and SQL

Fall 2025

Homework #2

Instructions

1. This is an individual assignment. You may not discuss your approach to solving these questions with anyone other than the instructor or TA.
2. Please include only your Student ID on the PDF submission.
3. The only allowed material is:
   a. Class notes
   b. Content posted on Canvas
   c. Textbook
4. You are not permitted to use other online resources
5. Due online, before the next lab.
6. There will be TA office hours. See the schedule on Canvas.

Background

The E-commerce Order Dataset provides structured information regarding orders, customers, products, payments, and order fulfillment for an online retail platform. The dataset is designed to facilitate comprehensive analysis of e-commerce operations, tracking customer behavior, order processing, and logistics.

This schema enables comprehensive analysis of customer purchasing behavior, seller performance, order fulfillment efficiency, and payment trends. Researchers and analysts can use this dataset to derive insights into e-commerce operations, optimizing business strategies and improving customer satisfaction.

Dataset Structure: The dataset consists of multiple relational tables, each focusing on different aspects of e-commerce transactions.

Orders (order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_timestamp, order_estimated_delivery_date**)**
This table records details related to customer purchases and order fulfillment.

- **order_id**: Unique identifier for each order.
- **customer_id**: Unique identifier for the customer placing the order.
- **order_status** : Status of the order (e.g., delivered, cancelled, processing).
- **order_purchase_timestamp** : Timestamp when the order was placed.
- **order_approved_at** : Timestamp when the order was approved.
- **order_delivered_timestamp** : Timestamp when the order was delivered.
- **order_estimated_delivery_date** : Expected delivery date communicated to the customer.

OrderItems (order_id, order_item_id, product_id, seller_id, price, shipping_charges)
Captures individual items within an order, linking products to transactions.

- **order_id**: Identifier linking items to an order.
- **order_item_id**: Line item number within the order.
- **product_id**: Unique identifier for the purchased product.
- **seller_id** : Unique identifier for the seller of the item.
- **price** : Price of the product.
- **shipping_charges** : Shipping cost associated with the item.

Customers (customer_id, customer_zip_code_prefix, customer_city, customer_state**)**
Stores information about customers registered on the platform.

- **customer_id** : Unique identifier for each customer.
- **customer_zip_code_prefix** : Zip code prefix of the customer.
- **customer_city** : City where the customer resides.
- **customer_state** : State where the customer resides.

Products (product_id, product_category_name, product_weight_g, product_length_cm, product_height_cm, product_width_cm)
Contains details of products available for sale.

- **product_id** : Unique identifier for each product.
- **product_category_name** : Name of the product category.
- **product_weight_g** : Weight of the product in grams.
- **product_length_cm** : Length of the product in centimeters.
- **product_height_cm** : Height of the product in centimeters.
- **product_width_cm** : Width of the product in centimeters.

Relations:

The dataset follows a relational structure, ensuring referential integrity among tables:
Orders (order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_timestamp, order_estimated_delivery_date**)**
OrderItems (order_id, order_item_id, product_id, seller_id, price, shipping_charges)
Customers (customer_id, customer_zip_code_prefix, customer_city, customer_state**)**
Products (product_id, product_category_name, product_weight_g, product_length_cm, product_height_cm, product_width_cm)

Database: The E/R Diagram for the database is below.

In Homework Part #1, submit answers to these questions online on Canvas by the due date. They are automatically graded.
In Homework part #2, include only answers to questions 7-14 in a PDF file and submit on the Part #2 page on Canvas.

For each question, submit your SQL code and a screenshot of the results. If the results
are too long, partial results are fine. Include relevant attributes for each result to explain that the result is correct. Do NOT include many unnecessary attributes.
*Do NOT use SELECT *.*

1- Select customers from the state "RN". Include customer ID, zip code, and customer city in the results.
2- For each state, count the number of customers. Include the state and the number of customers in your result. Sort the number of customers in descending order.
3- List all products from the category of health and beauty. Include product ID, product weight, and product height in your result. Sort by product weight in descending order.
4- For each product category, find the number of products and the sum of product weight. Sort your results by the number of products.
5- For toy products, find the average weight and the average height.
6- In the order items table with the price higher than $1000, list item ID, price, and shipping cost. Order by price in descending order.
7- List all orders that were delivered later than the estimated delivery date. In your result, include order ID, order estimated delivery date, and actual delivery date.
8- Find the total number of customers per state. Sort by the number of customers in descending order.
9- Count the total number of orders and the number of orders with missing shipping charges (NULL).
10- For each order status, compute the total number of orders, the earliest purchase date, and the latest purchase date.
11- Find the number of orders where the shipping charge is NULL.
12- Find the average product weight and dimensions for each product category and sort by average weight in descending order.
13- Find the highest-priced order item that does not have a NULL shipping charge. Only shows the top 5 rows and sorts by price in descending order.
14- How much time did you spend on this assignment?